indicated in the 2008 DSA, Papua New Guinea is vulnerable to terms of trade shocks.[3] However, unfavorable public debt dynamics did not materialize in 2009 as its terms of trade deteriorated during the first half of 2009 but recovered quickly with rising commodity prices.

The economic outlook of Papua New Guinea is expected to improve with the approval of two Liquefied Natural Gas (LNG) projects in December 2009. During the construction phase of the projects from 2010 to 2014, there will be increasing economic activities with rising imports mostly financed by FDI inflows. The near-term challenge will be to maintain macroeconomic stability with limited economic capacity. The LNG project is therefore included in the baseline scenario and Box 1 summarizes the medium-term macroeconomic framework underlying the DSA.

III. External Debt Sustainability Analysis

The baseline scenario indicates that all PPG external debt and debt service ratios stay well below the policy-dependent debt burden thresholds. However, the vulnerability of the external debt sustainability remains if export value growth slows down in combination with other shocks. The main results of the external DSA are the following:

- **The baseline scenario shows a slight increase in external debt burden in the medium term before declining in the long run** (Table 3 and the text table). This is due to expected increase in disbursements of approved AsDB loans in the pipeline. Since additional financing needs for the LNG project and the current account deficit would be met with projected large FDI inflows, the PV of PPG external debt over GDP would remain around 11 percent in the medium term, which is below the 30 percent of threshold. The PVs of PPG external debt-to-exports and debt-to-revenue ratios are expected to stay well below the indicative threshold levels over the projection period.

- **However, the debt burden is expected to breach the indicative thresholds under some stress tests** (Table 4 and Figure 1). External debt sustainability is most vulnerable to shocks from lower export growth (Table 4). A decline in export value growth in 2010–11, by one standard deviation below its historical average, would raise the PV of public external debt-to-GDP ratio to 37 percentage points over the medium term, and only fall below the 30 percent threshold after 2018. Other stress tests including lower FDI inflows and a combination of several shocks above also generate similar debt patterns which move to alarming levels.[4]

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[3] See Papua New Guinea: 2008 Article IV Consultation, IMF Staff Report 09/112.

[4] Bound test B5, which combines four different shocks with smaller standard deviations than those shocks in tests B1–B4, also shows the similar pattern of the debt dynamics with the shock from the export growth dominates.

- **The scenarios indicated by stress tests on external debt burden are not trivial and cannot be ruled out given the uncertainty ahead.** Export demand could remain weak if the recovery of the global economy is sluggish and a "double-dip" of commodity prices happens. Moreover, the negotiation between the government and local communities on land issues is ongoing and the construction of the LNG project would not be fully launched before these issues are settled.

Text Table: Policy-based PPG External Debt Burden Thresholds for PNG

	Thresholds	2009	PNG's ratios 2010–15	2016–30
PV of debt in percent of				
GDP	30	12.2	10.8	9.4
Exports	100	19.8	15.3	15.0
Revenues	200	52.4	41.0	30.7
Debt service in percent of				
Exports	15	2.7	1.6	1.0
Revenues	25	7.2	4.4	2.1

IV. Public Debt Sustainability Analysis[5]

With a prudent fiscal policy and improved economic outlook, the overall public debt dynamics for Papua New Guinea is favorable. During previous years of rising commodity prices, windfall revenues were largely saved and public debt reduced. The public debt burden is forecasted to decrease over the projection period under the baseline. Meanwhile, compared to the 2008 Article IV DSA, given the LNG project now included in the baseline, it indicates Papua New Guinea remains vulnerable to external shocks as well as domestic ones during the construction period of the project. In addition, the inclusion of several contingent domestic liabilities shows potential risks to long-run fiscal sustainability.

- **The baseline scenario projects a gradual decline in the public sector debt ratio over the medium term.** The public debt ratio is projected to decline to 23.8 percent of GDP by 2015 (Table 1). The projected improvement results mainly from projected rapid economic growth, large FDI inflows associated with the LNG project, a continued prudent fiscal policy. The public debt is projected to decline further to about 14 percent by 2030.

- **The alternative scenarios and bound tests for public debt sustainability show that the projected debt path is particularly sensitive to changes in real GDP**

[5] Public debt includes domestic central government debt and external public and publicly guaranteed debt.

growth (Table 2 and Figure 2). A negative shock to real GDP growth[6] in 2011–12 would increase the present value (PV) of the public debt-to-GDP ratio close to 80 percent in 2030. This shock can be again related with the progress of the LNG construction given its significant contribution to the economic growth over the medium term. Moreover, if a primary deficit of 5.8 percent in 2009 is assumed to maintain for the projection period, the PV of the public debt-to-GDP ratio would be above 200 percent in 2030. While the case of a persistent primary deficit is unlikely to happen (in view of the unusual spending from the government trust account in 2009), this highlights the need for the government to ensure the project is moving forward as planned while protecting the priority spending in the event of a growth slowdown. In addition, as discussed in the previous section, a decline in the export growth would also exacerbate public debt burden.

- **Other country-specific liabilities from several potential sources may also undermine fiscal sustainability.** Government unfunded superannuation liabilities, estimated to be about 9 percent of GDP by end-2009, will remain sizable though the government has made partial repayment recently. The contingent liabilities related with the LNG project are estimated to be, in PV term, about 20 percent of GDP in 2014.[7] Other liabilities, including borrowings by SOEs[8] and the government's commitments under the memorandum of agreement with landowners, are also estimated to be significant though detailed information on the magnitude and timing is limited. These factors, as shown in the alternative scenario[9] (Table 2 and Figure 2), would result in unfavorable debt dynamics and add additional risks to public debt sustainability.

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[6] The GDP growth shock in 2011–12 is defined as the historical average of GDP growth rate minus one standard deviation. Another permanent GDP growth shock, which will bring the Debt-to-GDP ratio above 70 percent in 2030, is defined as the baseline GDP growth rate minus one standard deviation divided by the square root of the length of the projection period.

[7] These liabilities are arising from the issue of exchangeable bonds to finance the state's equity participation (in 2009) and the state's completion guarantee for debt financing (by 2014). Under this scheme, the creditor acquires exchangeable bonds, amounting to US$1.1 billion, with the option to exchange the bonds for equities in the Oil Search company. Under the completion guarantee, if the project fails, the government may have to pay up to US$2 billion to the creditor. Other liabilities including the settlements with landowners (negotiation is ongoing) may also be significant.

[8] From 2008 Article IV DSA estimates, SOEs related contingent liabilities are about 2.5 percent of GDP when we consider borrowings, various provisions, and other liabilities based on end-2007 financial statements of nine SOEs. However, there is no information on the off-balance sheet liabilities which also may be significant.

[9] The alternative scenario (All Other Liabilities: Pension, SOEs, and LNG contingent liabilities) assumes that pension and SOEs liabilities are to be covered in the 2011 budget (total about 10 percent of GDP) and all LNG contingent liabilities will materialize in 2014 (about 20 percent of GDP).

BOX 1. MACROECONOMIC ASSUMPTIONS UNDERLYING THE DSA

- **Real GDP growth** is projected to be at 5.2 percent on average over the medium term, above the historical average of 2.9 percent, and slow to 3 percent in the long run. In the medium term rapid growth in the nonmineral sector, including in agriculture, as well as several mining projects coming on stream will offset the decline in petroleum and copper production and exports.
- **Inflation** is projected to rise in the medium term due to increasing domestic demand and near-term capacity constraints, while stabilizing at around 3 percent in the long run.
- The **two LNG projects** have a construction period of 5 years each, from 2010/11 to 2014/15. Production and exports are expected to commence in the second half of 2014, with a maximum capacity of about 60 million barrels of oil equivalent. The peak impact of the projects is estimated to be an increase of 25 to 30 percent in annual GDP growth. These 30-year projects are estimated to increase annual GDP growth, through both direct and indirect channels, by about 15 to 20 percent on average. Accounting for the substantial income outflows, the projects are expected to increase annual GNI by about 9 percent.
- The **current account** (including grants) is estimated to be in deficit in 2009 and the deficit is projected to remain over the medium term until 2013, primarily reflecting the strong import growth related with the construction phase of the LNG project. FDI inflows from foreign shareholders are projected to increase substantially given the financing scheme of the project.
- The **grant element of loans** is expected to decline. As per capita income rises, the share of external financing provided on concessional terms is expected to decline gradually over the projection period.[1]
- The **fiscal balance** is estimated to turn into a deficit[2] in 2009 due to falling mineral revenues and the spending of preceding years' mineral windfall revenues. The deficit is financed by a rundown of resources held in trust accounts. Over the medium term it is projected to remain balanced.

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[1] Grant-equivalent financing (in percent of GDP) increases slightly in the near term before it declines in the long run, due to the expected large disbursement from the AsDB in the pipeline.

[2] The deficit is estimated by Fund staff. The official 2009 budget outcome does not include the large amount of spending from the government trust account held outside the budget. The trust accounts consist of saving from mineral windfall revenues in past years to address development and debt reduction needs. However, since 2008 the government has relaxed constraints on trust account spending. Trust spending was the main factor shifting the fiscal balance from a surplus of 2.5 percent of GDP in 2008 to a deficit of almost 8 percent of GDP in 2009. The result was a fiscal impulse of almost 13 percent of GDP.

V. CONCLUSION

It is the staffs' view that Papua New Guinea should be considered at moderate risk of debt stress based on both external debt burden indicators and stress tests on public debt pattern. Although the baseline scenario indicates that all external PPG debt burden indicators are below their indicative thresholds, the projected PPG debt paths shows significant vulnerability against shocks from slower export and economic growth. The inclusion of the LNG contingent liability as well as other domestic liabilities would also add concerns to long-term fiscal sustainability.

Enhanced debt management with adherence to the Medium-Term Fiscal Strategy (MTFS) would be required to offset the underlying risks in public debt sustainability. The staff encourages the authorities to improve implementation of the MTFS during the construction phase of the LNG project to guard against unfavorable public debt dynamics from vulnerabilities such as those arising from potential LNG project delays, lower commodity prices, and weaker external demand. The staff also encourages the authorities to consider the establishment of a Sovereign Wealth Fund (SWF) to manage revenues from those projects more efficiently and to further strengthen debt management capacity.

Table 1.Papua New Guinea: Public Sector Debt Sustainability Framework, Baseline Scenario, 2007-2030
(In percent of GDP, unless otherwise indicated)

	Actual					Estimate	Projections								
	2007	2008	2009	Average	Standard Deviation	2010	2011	2012	2013	2014	2015	2010-15 Average	2020	2030	2016-30 Average
Public sector debt 1/	34.1	31.7	32.0			28.8	26.5	26.7	26.4	22.2	23.8		21.2	14.5	
o/w foreign-currency denominated	17.1	13.2	12.8			11.9	11.8	12.6	13.5	12.8	13.7		13.0	9.1	
Change in public sector debt	-5.6	-2.4	0.4			-3.3	-2.2	0.1	-0.2	-4.2	1.6		-0.6	-0.6	
Identified debt-creating flows	-14.2	-10.8	7.3			-1.2	-2.4	-1.5	-0.9	-7.8	-3.1		-1.8	-0.9	
Primary deficit	-11.0	-7.2	5.8	-3.8	4.8	0.2	-2.5	-2.8	-2.0	-4.3	-3.4	-2.5	-1.6	-0.7	-1.5
Revenue and grants	37.3	34.8	28.3			27.9	29.8	30.1	29.4	28.2	27.1		28.8	32.1	
of which: grants	3.8	5.1	4.7			2.0	2.5	2.4	2.2	1.8	1.9		1.5	0.8	
Primary (noninterest) expenditure	26.3	27.6	34.1			28.1	27.3	27.3	27.3	23.9	23.7		27.3	31.3	
Automatic debt dynamics	-3.2	-3.6	1.5			-1.4	0.1	1.3	1.2	-3.5	0.3		-0.2	-0.2	
Contribution from interest rate/growth differential	-1.4	-1.8	0.7			-2.0	-0.5	0.6	0.7	-2.4	-0.2		-0.1	-0.1	
of which: contribution from average real interest rate	1.2	0.4	2.0			0.4	1.0	1.4	1.2	-0.5	0.8		0.5	0.4	
of which: contribution from real GDP growth	-2.6	-2.1	-1.4			-2.4	-1.5	-0.8	-0.5	-2.0	-1.0		-0.6	-0.4	
Contribution from real exchange rate depreciation	-1.7	-1.9	0.8			0.5	0.7	0.7	0.5	-1.1	0.5		...	...	
Other identified debt-creating flows	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Privatization receipts (negative)	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Recognition of implicit or contingent liabilities	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Debt relief (HIPC and other)	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Other (specify, e.g. bank recapitalization)	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Residual, including asset changes	8.6	8.4	-7.0			-2.0	0.1	1.6	0.6	3.6	4.7		1.2	0.3	
Other Sustainability Indicators															
PV of public sector debt	16.9	18.5	31.4			28.0	25.4	25.0	24.1	19.6	20.7		18.3	13.2	
o/w foreign-currency denominated	0.0	0.0	12.2			11.1	10.7	10.9	11.2	10.2	10.6		10.1	7.8	
o/w external	...	...	12.2			11.1	10.7	10.9	11.2	10.2	10.6		10.1	7.8	
PV of contingent liabilities (not included in public sector debt)	...	...	...			...	...	...	...	...	...		...	...	
Gross financing need 2/	6.3	4.6	20.6			11.0	7.6	6.6	6.7	2.8	3.0		3.3	3.3	
PV of public sector debt-to-revenue and grants ratio (in percent)	45.4	53.2	111.1			100.4	85.4	82.9	82.0	69.6	76.6		63.4	41.1	
PV of public sector debt-to-revenue ratio (in percent)	50.7	62.4	133.4			108.1	93.0	89.9	88.6	74.3	82.4		66.8	42.2	
o/w external 3/	…	…	51.9			42.9	39.3	39.4	41.2	38.5	42.4		36.8	24.9	
Debt service-to-revenue and grants ratio (in percent) 4/	34.0	24.4	30.3			17.0	17.0	17.9	18.5	16.2	15.7		10.0	8.5	
Debt service-to-revenue ratio (in percent) 4/	37.9	28.7	36.3			18.3	18.5	19.4	20.0	17.3	16.9		10.6	8.8	
Primary deficit that stabilizes the debt-to-GDP ratio	-5.5	-4.8	5.5			3.5	-0.3	-2.9	-1.8	-0.1	-5.0		-0.9	-0.1	
Key macroeconomic and fiscal assumptions															
Real GDP growth (in percent)	7.2	6.7	4.5	2.9	3.0	8.0	5.5	3.0	2.0	8.0	5.0	5.2	3.0	3.0	3.0
Average nominal interest rate on forex debt (in percent)	3.2	2.7	2.6	3.2	0.4	2.8	2.8	2.8	2.6	2.3	2.2	2.6	1.8	2.1	1.9
Average real interest rate on domestic debt (in percent)	5.0	0.7	10.9	3.9	5.8	1.8	5.6	9.2	8.3	-4.3	9.0	4.9	6.5	6.5	6.5
Real exchange rate depreciation (in percent, + indicates depreciation)	-8.5	-11.4	6.5	-3.9	10.5	4.4	...	...	...	...	...	...	...	...	...
Inflation rate (GDP deflator, in percent)	3.8	7.9	-3.6	6.5	5.0	4.7	2.2	0.2	1.4	14.8	1.1	4.1	3.0	3.0	3.0
Growth of real primary spending (deflated by GDP deflator, in percent)	0.0	0.1	0.3	0.1	0.1	-0.1	0.0	0.0	0.0	-0.1	0.0	0.0	0.0	0.1	0.0
Grant element of new external borrowing (in percent)	...	...	...	…	…	42.2	41.8	41.5	41.2	34.3	33.9	39.2	20.2	11.5	...

Sources: Country authorities; and staff estimates and projections.
1/ Public sector includes central government and nonfinancial public sector. Gross debt is used.
2/ Gross financing need is defined as the primary deficit plus debt service plus the stock of short-term debt at the end of the last period.
3/ Revenues excluding grants.
4/ Debt service is defined as the sum of interest and amortization of medium and long-term debt.
5/ Historical averages and standard deviations are generally derived over the past 10 years, subject to data availability.

Table 2.Papua New Guinea: Sensitivity Analysis for Key Indicators of Public Debt 2010-2030

	Projections							
	2010	2011	2012	2013	2014	2015	2020	2030
PV of Debt-to-GDP Ratio								
Baseline	28	25	25	24	20	21	18	13
A. Alternative scenarios								
A1. Real GDP growth and primary balance are at historical averages	28	25	23	20	18	19	6	-27
A2. Primary balance is unchanged from 2010	28	34	43	51	56	67	112	210
A3. Permanently lower GDP growth 1/	28	26	26	26	22	24	30	65
A4. All Other Liabilities (Pension, SOEs, and LNG contingent liabilities)	28	36	36	36	51	54	57	66
B. Bound tests								
B1. Real GDP growth is at historical average minus one standard deviations in 2011-2012	28	28	32	33	31	34	46	76
B2. Primary balance is at historical average minus one standard deviations in 2011-2012	28	29	33	32	27	29	27	24
B3. Combination of B1-B2 using one half standard deviation shocks	28	28	29	30	27	30	36	54
B4. One-time 30 percent real depreciation in 2011	28	31	30	29	24	25	23	21
B5. 10 percent of GDP increase in other debt-creating flows in 2011	28	36	36	35	30	32	30	28
PV of Debt-to-Revenue Ratio								
Baseline	100	85	83	82	70	77	63	41
A. Alternative scenarios								
A1. Real GDP growth and primary balance are at historical averages	100	83	78	70	63	71	22	-85
A2. Primary balance is unchanged from 2010	100	115	143	174	199	249	387	654
A3. Permanently lower GDP growth 1/	100	86	86	87	78	89	105	202
A4. All Other Liabilities (Pension, SOEs, and LNG contingent liabilities)	100	121	120	123	181	199	199	205
B. Bound tests								
B1. Real GDP growth is at historical average minus one standard deviations in 2011-2012	100	95	104	113	108	126	159	237
B2. Primary balance is at historical average minus one standard deviations in 2011-2012	100	98	109	110	97	106	94	74
B3. Combination of B1-B2 using one half standard deviation shocks	100	93	97	103	95	109	126	168
B4. One-time 30 percent real depreciation in 2011	100	103	100	99	85	92	79	65
B5. 10 percent of GDP increase in other debt-creating flows in 2011	100	121	119	120	108	117	105	86
Debt Service-to-Revenue Ratio								
Baseline	17	17	18	18	16	16	10	9
A. Alternative scenarios								
A1. Real GDP growth and primary balance are at historical averages	17	17	18	18	14	14	2	-18
A2. Primary balance is unchanged from 2010	17	17	19	27	40	39	79	147
A3. Permanently lower GDP growth 1/	17	17	18	19	17	17	17	42
A4. All Other Liabilities (Pension, SOEs, and LNG contingent liabilities)	17	17	20	27	24	42	40	42
B. Bound tests								
B1. Real GDP growth is at historical average minus one standard deviations in 2011-2012	17	18	20	22	23	23	29	52
B2. Primary balance is at historical average minus one standard deviations in 2011-2012	17	17	19	22	25	22	17	16
B3. Combination of B1-B2 using one half standard deviation shocks	17	18	19	21	21	21	23	37
B4. One-time 30 percent real depreciation in 2011	17	18	20	21	19	19	15	16
B5. 10 percent of GDP increase in other debt-creating flows in 2011	17	17	20	27	28	25	20	19

Sources: Country authorities; and staff estimates and projections.
1/ Assumes that real GDP growth is at baseline minus one standard deviation divided by the square root of the length of the projection period.
2/ Revenues are defined inclusive of grants.

Table 3. External Debt Sustainability Framework, Baseline Scenario, 2007-2030 1/
(In percent of GDP, unless otherwise indicated)

	Actual			Historical 0 Average 0	Standard Deviation	Projections									
	2007	2008	2009			2010	2011	2012	2013	2014	2015	2010-2015 Average	2020	2030	2016-203(Average
External debt (nominal) 1/	**31.2**	**25.5**	**26.9**			**25.8**	**26.7**	**29.0**	**30.2**	**27.1**	**28.1**		**27.4**	**23.4**	
o/w public and publicly guaranteed (PPG)	17.1	13.2	12.8			11.9	11.8	12.6	13.5	12.8	13.7		13.0	9.1	
Change in external debt	-7.9	-5.8	1.4			-1.1	0.9	2.2	1.3	-3.1	1.0		-0.3	-0.4	
Identified net debt-creating flows	-15.6	-16.2	-2.5			1.3	2.3	2.5	1.8	-9.2	-10.3		-7.5	-3.3	
Non-interest current account deficit	**-5.6**	**-11.4**	**5.4**	**-6.3**	**5.2**	**14.7**	**17.0**	**14.8**	**10.0**	**-4.8**	**-7.0**		**-5.4**	**-2.3**	-4.3
Deficit in balance of goods and services	-7.3	-16.0	1.6			11.2	13.8	11.8	7.2	-13.6	-15.7		-12.6	-6.9	
Exports	80.9	77.2	61.9			68.7	75.1	74.3	70.8	68.1	67.6		64.3	59.0	
Imports	73.6	61.1	63.6			79.9	88.9	86.2	77.9	54.5	51.9		51.7	52.1	
Net current transfers (negative = inflow)	-5.5	-2.0	-1.2	-4.8	2.2	-1.6	-1.3	-1.1	-1.1	-0.8	-0.7		-0.5	-0.2	-0.4
o/w official	-4.9	-1.6	-2.2			-2.5	-2.3	-2.2	-2.2	-2.1	-2.0		-1.6	-0.9	
Other current account flows (negative = net inflow)	7.2	6.7	4.9			5.1	4.5	4.0	3.9	9.6	9.4		7.7	4.7	
Net FDI (negative = inflow)	**-7.3**	**0.4**	**-9.7**	**-3.1**	**3.1**	**-12.9**	**-14.8**	**-13.1**	**-9.3**	**-3.8**	**-3.5**		**-2.7**	**-1.6**	-2.4
Endogenous debt dynamics 2/	**-2.7**	**-5.2**	**1.8**			**-0.5**	**0.1**	**0.8**	**1.1**	**-0.6**	**0.1**		**0.6**	**0.7**	
Contribution from nominal interest rate	2.3	1.3	1.5			1.4	1.5	1.7	1.7	1.4	1.4		1.4	1.3	
Contribution from real GDP growth	-2.4	-1.6	-1.2			-1.9	-1.4	-0.8	-0.6	-2.0	-1.3		-0.8	-0.7	
Contribution from price and exchange rate changes	-2.6	-4.9	1.5			…	…	…	…	…	…		…	…	
Residual (3-4) 3/	**7.7**	**10.4**	**4.0**			**-2.4**	**-1.4**	**-0.3**	**-0.5**	**6.1**	**11.4**		**7.3**	**2.9**	
o/w exceptional financing	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
PV of external debt 4/	...	...	26.3			25.0	25.6	27.3	27.9	24.5	25.0		24.4	22.1	
In percent of exports	...	...	42.5			36.5	34.1	36.7	39.4	36.0	37.0		38.0	37.5	
PV of PPG external debt	**...**	**...**	**12.2**			**11.1**	**10.7**	**10.9**	**11.2**	**10.2**	**10.6**		**10.1**	**7.8**	
In percent of exports	**...**	**...**	**19.8**			**16.2**	**14.3**	**14.7**	**15.8**	**14.9**	**15.8**		**15.7**	**13.2**	
In percent of government revenues	**...**	**...**	**51.9**			**42.9**	**39.3**	**39.4**	**41.2**	**38.5**	**42.4**		**36.8**	**24.9**	
Debt service-to-exports ratio (in percent)	**8.3**	**5.1**	**7.0**			**5.7**	**5.2**	**5.4**	**7.0**	**6.0**	**6.1**		**5.6**	**6.8**	
PPG debt service-to-exports ratio (in percent)	**4.1**	**2.0**	**2.7**			**2.0**	**1.7**	**1.6**	**1.7**	**1.4**	**1.4**		**0.9**	**1.0**	
PPG debt service-to-revenue ratio (in percent)	**9.8**	**5.2**	**7.1**			**5.2**	**4.6**	**4.4**	**4.4**	**3.7**	**3.8**		**2.2**	**1.8**	
Total gross financing need (Billions of U.S. dollars)	-0.4	-0.6	0.0			0.5	0.5	0.5	0.5	-0.5	-0.7		-0.6	0.0	
Non-interest current account deficit that stabilizes debt ratio	2.4	-5.7	3.9			15.8	16.1	12.6	8.7	-1.7	-8.0		-5.1	-2.0	
Key macroeconomic assumptions															
Real GDP growth (in percent)	7.2	6.7	4.5	2.9	3.0	8.0	5.5	3.0	2.0	8.0	5.0	5.2	3.0	3.0	3.0
GDP deflator in US dollar terms (change in percent)	7.0	18.5	-5.6	6.0	10.1	3.2	-4.0	-4.7	-2.2	11.4	-1.8	0.3	3.0	3.0	2.9
Effective interest rate (percent) 5/	6.6	5.4	5.7	4.8	1.4	5.9	6.0	6.1	5.7	5.6	5.4	5.8	5.3	6.0	5.5
Growth of exports of G&S (US dollar terms, in percent)	11.5	20.5	-20.8	9.7	19.7	23.6	10.8	-2.9	-5.1	15.9	2.3	7.4	4.8	5.6	5.0
Growth of imports of G&S (US dollar terms, in percent)	30.7	4.9	2.7	12.2	19.1	40.1	12.7	-4.9	-9.8	-15.8	-1.9	3.4	5.9	6.3	6.0
Grant element of new public sector borrowing (in percent)	...	...	...	...	...	42.2	41.8	41.5	41.2	34.3	33.9	39.2	20.2	11.5	18.5
Government revenues (excluding grants, in percent of GDP)	33.4	29.6	23.6			25.9	27.3	27.8	27.2	26.4	25.1		27.3	31.2	28.5
Aid flows (in Billions of US dollars) 7/	0.3	0.4	0.4			0.2	0.3	0.3	0.3	0.4	0.4		0.3	0.2	
o/w Grants	0.2	0.4	0.4			0.2	0.2	0.2	0.2	0.2	0.2		0.2	0.2	
o/w Concessional loans	0.0	0.0	0.1			0.1	0.1	0.1	0.2	0.2	0.2		0.0	0.0	
Grant-equivalent financing (in percent of GDP) 8/	...	...	...			2.3	2.9	3.0	2.9	2.6	2.6		1.7	0.9	1.5
Grant-equivalent financing (in percent of external financing) 8/	...	...	...			84.2	81.8	77.0	73.2	63.8	65.8		70.9	64.9	69.3
Memorandum items:															
Nominal GDP (Billions of US dollars)	6.3	8.0	7.9			8.8	8.9	8.8	8.7	10.5	10.8		14.3	25.9	
PV of PPG external debt (in Billions of US dollars)			1.0			0.9	0.9	0.9	1.0	1.1	1.1		1.4	2.0	
Gross remittances (Billions of US dollars)	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
PV of PPG external debt (in percent of GDP + remittances)	...	...	12.2			11.1	10.7	10.9	11.2	10.2	10.6		10.1	7.8	
PV of PPG external debt (in percent of exports + remittances)	...	...	19.8			16.2	14.3	14.7	15.8	14.9	15.8		15.7	13.2	
Debt service of PPG external debt (in percent of exports + remittances)	...	...	2.7			2.0	1.7	1.6	1.7	1.4	1.4		0.9	1.0	

Sources: Country authorities; and staff estimates and projections.

1/ Includes both public and private sector external debt.
2/ Derived as [r - g - ρ(1+g)]/(1+g+ρ+gρ) times previous period debt ratio, with r = nominal interest rate; g = real GDP growth rate, and ρ = growth rate of GDP deflator in U.S. dollar terms.
3/ Includes exceptional financing (i.e., changes in arrears and debt relief); changes in gross foreign assets; and valuation adjustments. For projections also includes contribution from price and exchange rate changes.
4/ Assumes that PV of private sector debt is equivalent to its face value.
5/ Current-year interest payments divided by previous period debt stock.
6/ Historical averages and standard deviations are generally derived over the past 10 years, subject to data availability.
7/ Defined as grants, concessional loans, and debt relief.
8/ Grant-equivalent financing includes grants provided directly to the government and through new borrowing (difference between the face value and the PV of new debt).

Table 4a.Papua New Guinea: Sensitivity Analysis for Key Indicators of Public and Publicly Guaranteed External Debt, 2010-2030
(In percent)

	Projections							
	2010	2011	2012	2013	2014	2015	**2020**	2030
PV of debt-to GDP ratio								
Baseline	11	11	11	11	10	11	**10**	8
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2010-2030 1/	11	1	-8	-15	-15	-13	**-8**	-27
A2. New public sector loans on less favorable terms in 2010-2030 2	11	11	11	12	12	13	**13**	12
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2011-2012	11	11	12	12	11	11	**11**	8
B2. Export value growth at historical average minus one standard deviation in 2011-2012 3/	11	21	36	37	33	33	**28**	15
B3. US dollar GDP deflator at historical average minus one standard deviation in 2011-2012	11	10	11	11	10	10	**10**	8
B4. Net non-debt creating flows at historical average minus one standard deviation in 2011-2012 4/	11	21	31	32	28	28	**24**	14
B5. Combination of B1-B4 using one-half standard deviation shocks	11	24	34	36	31	32	**27**	15
B6. One-time 30 percent nominal depreciation relative to the baseline in 2011 5/	11	15	16	16	14	15	**15**	11
PV of debt-to-exports ratio								
Baseline	16	14	15	16	15	16	**16**	13
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2010-2030 1/	16	1	-11	-22	-21	-19	**-13**	-45
A2. New public sector loans on less favorable terms in 2010-2030 2	16	14	15	17	17	19	**21**	20
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2011-2012	16	14	14	16	15	15	**16**	13
B2. Export value growth at historical average minus one standard deviation in 2011-2012 3/	16	35	65	70	64	65	**58**	35
B3. US dollar GDP deflator at historical average minus one standard deviation in 2011-2012	16	14	14	16	15	15	**16**	13
B4. Net non-debt creating flows at historical average minus one standard deviation in 2011-2012 4/	16	28	41	45	41	42	**37**	23
B5. Combination of B1-B4 using one-half standard deviation shocks	16	36	53	57	52	53	**47**	28
B6. One-time 30 percent nominal depreciation relative to the baseline in 2011 5/	16	14	14	16	15	15	**16**	13
PV of debt-to-revenue ratio								
Baseline	43	39	39	41	38	42	**37**	25
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2010-2030 1/	43	3	-29	-56	-55	-50	**-31**	-86
A2. New public sector loans on less favorable terms in 2010-2030 2	43	39	41	45	44	50	**49**	38
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2011-2012	43	40	42	44	41	45	**40**	27
B2. Export value growth at historical average minus one standard deviation in 2011-2012 3/	43	78	130	138	124	132	**103**	49
B3. US dollar GDP deflator at historical average minus one standard deviation in 2011-2012	43	38	38	40	38	41	**36**	25
B4. Net non-debt creating flows at historical average minus one standard deviation in 2011-2012 4/	43	76	110	117	105	112	**88**	44
B5. Combination of B1-B4 using one-half standard deviation shocks	43	89	124	131	118	126	**98**	46
B6. One-time 30 percent nominal depreciation relative to the baseline in 2011 5/	43	55	56	59	55	60	**53**	36

Table 4b.Papua New Guinea: Sensitivity Analysis for Key Indicators of Public and Publicly Guaranteed External Debt, 2010-2030 (cont.)
(In percent)

Debt service-to-exports ratio

Baseline	2	2	2	2	1	1	**1**	1
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2010-2030 1/	2	2	1	1	0	0	**-1**	-2
A2. New public sector loans on less favorable terms in 2010-2030 2	2	2	2	2	2	2	**1**	2
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2011-2012	2	2	2	2	1	1	**1**	1
B2. Export value growth at historical average minus one standard deviation in 2011-2012 3/	2	2	3	4	3	3	**4**	3
B3. US dollar GDP deflator at historical average minus one standard deviation in 2011-2012	2	2	2	2	1	1	**1**	1
B4. Net non-debt creating flow s at historical average minus one standard deviation in 2011-2012 4/	2	2	2	2	2	2	**3**	2
B5. Combination of B1-B4 using one-half standard deviation shocks	2	2	2	3	2	2	**3**	2
B6. One-time 30 percent nominal depreciation relative to the baseline in 2011 5/	2	2	2	2	1	1	**1**	1

Debt service-to-revenue ratio

Baseline	5	5	4	4	4	4	**2**	2
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2010-2030 1/	5	4	3	2	1	1	**-2**	-3
A2. New public sector loans on less favorable terms in 2010-2030 2	5	5	4	5	4	4	**3**	3
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2011-2012	5	5	5	5	4	4	**2**	2
B2. Export value growth at historical average minus one standard deviation in 2011-2012 3/	5	5	5	7	6	6	**7**	4
B3. US dollar GDP deflator at historical average minus one standard deviation in 2011-2012	5	5	4	4	4	4	**2**	2
B4. Net non-debt creating flow s at historical average minus one standard deviation in 2011-2012 4/	5	5	5	6	5	6	**6**	4
B5. Combination of B1-B4 using one-half standard deviation shocks	5	5	5	7	6	6	**7**	4
B6. One-time 30 percent nominal depreciation relative to the baseline in 2011 5/	5	7	6	6	5	6	**3**	3
Memorandum item:								
Grant element assumed on residual financing (i.e., financing required above baseline) 6/	23	23	23	23	23	23	**23**	23

Sources: Country authorities; and staff estimates and projections.

1/ Variables include real GDP grow th, grow th of GDP deflator (in U.S. dollar terms), non-interest current account in percent of GDP, and non-debt creating flow s.

2/ Assumes that the interest rate on new borrow ing is by 2 percentage points higher than in the baseline., w hile grace and maturity periods are the same as in the baseline.

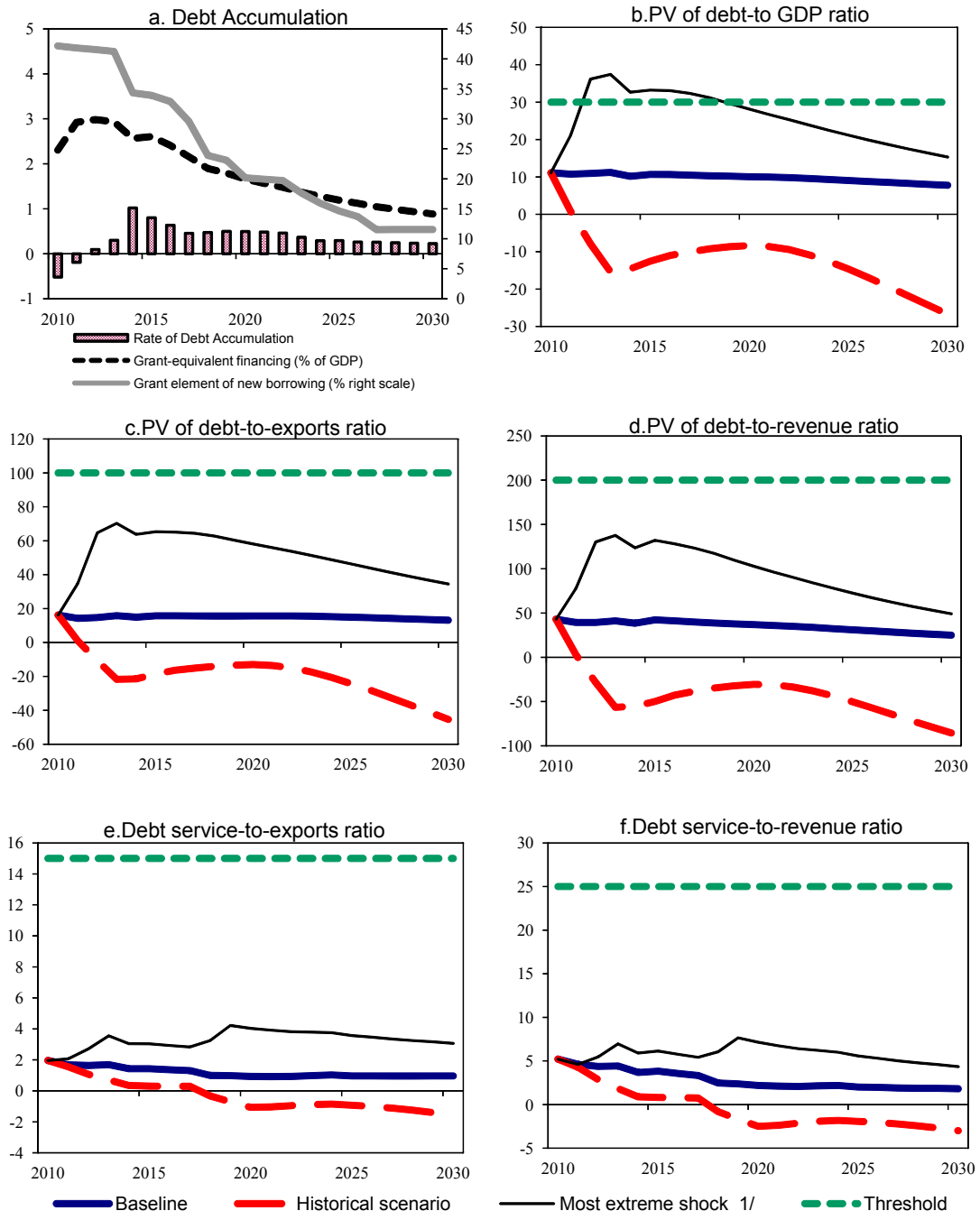
3/ Exports values are assumed to remain permanently at the low er level, but the current account as a share of GDP is assumed to return to its baseline level after the shock (implicitly assuming an offsetting adjustment in import levels).

4/ Includes official and private transfers and FDI.

5/ Depreciation is defined as percentage decline in dollar/local currency rate, such that it never exceeds 100 percent.

6/ Applies to all stress scenarios except for A2 (less favorable financing) in w hich the terms on all new financing are as specified in footnote 2.

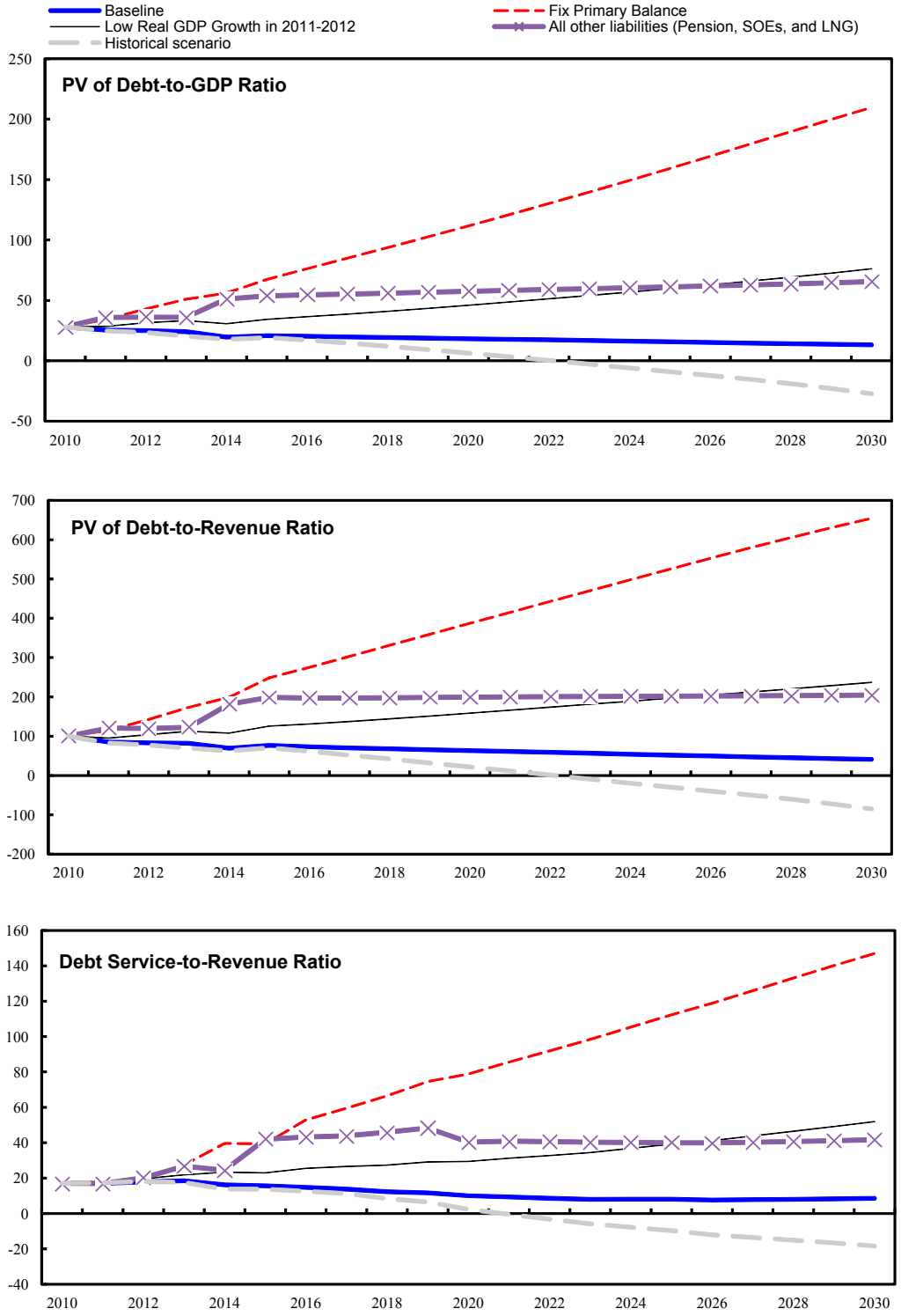
Figure 1. Papua New Guinea: Indicators of Public and Publicly Guaranteed External Debt under Alternatives Scenarios, 2010-2030 1/



Sources: Country authorities; and staff estimates and projections.

1/ The most extreme stress test is the test that yields the highest ratio in 2020. In figure b. it corresponds to a Exports shock; in c. to a Exports shock; in d. to a Exports shock; in e. to a Exports shock and in figure f. to a Exports shock

Figure 2.Papua New Guinea: Indicators of Public Debt Under Alternative Scenarios, 2010-2030 1/



Sources: Country authorities; and staff estimates and projections.

1/ Revenues are defined inclusive of grants.